

Dogs and relatives

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Carnivora
FAMILY	Canidae
SPECIES	35

Members of the dog family—dogs, wolves, coyotes, jackals, and foxes—are collectively described as canids. They are known for great endurance (rather than sudden bursts of speed) and for opportunistic and adaptable behavior. Dogs are characterized by a slender build, long legs, and a long, bushy tail. Wild canids generally inhabit open grassland and forested habitats the world over—the dingo was introduced to Australia by humans about 4,000 years ago, and feral domestic dogs are found in many isolated areas.

Anatomy

Canids have a muscular, deep-chested body covered with a fur coat that is usually uniformly colored or speckled. The lower limbs are modified, with fused wrist bones, and a separate ulna and radius that lock together. This prevents the rotation of the lower limbs. There are 4 digits on the back feet and 5 on the front feet, and each digit has a hard pad. The claws are short, nonretractable, and blunt (other carnivores have sharp claws). Canids also have long jaws, long, fanglike canines (for stabbing prey), and well-developed carnassials (the shearing teeth at the back of the jaws). Canids track their prey by scent, and the long, pointed muzzle houses

large olfactory organs. Hearing is also acute, and the ears are large, erect, and usually pointed. Sight is less important, but is still well developed.

Social groups

Smaller species, which usually feed mainly on small rodents and insects, tend to have a flexible social organization but often live either in pairs (for example, jackals) or alone (for example, foxes). However, larger species, such as the wolf and the African wild dog, live in social groups called packs. These packs, which consist of a dominant pair and their offspring, occupy and defend

territories, which they mark with urine. The young in a pack are of different ages because older offspring remain in the group for some years, and may help rear new young. Only the dominant pair breed, and the female digs a den in which to give birth. The pack will often perform a bonding ritual, which involves mutual licking, whining, and tail wagging. When hunting, the usual tactic is to track a herd of deer or antelopes (for example) and then cooperatively maneuver so as to separate one animal. This individual is then run down and bitten by pack members until it falls, exhausted. On returning to the den, the hunters regurgitate meat for the cubs to eat.

Canids and people

Throughout history, canids have proved useful to humankind in many ways. Wild canids are, for example, important controllers of rodent populations, which can spiral quickly if left unchecked. Furthermore, the domestic dog—which descended

from the wolf over 10,000 years ago—has always played an important role in a number of human activities. From the tiny chihuahua to the huge St. Bernard (the domestic dog displays more variation between types than any other domestic animal), there are breeds specialized for hunting, herding, guarding, performing, carrying or dragging loads, and companionship.

However, many canid species are considered pests. The wolf, for example, has been hunted and persecuted as a killer of livestock. As a result, this species is now rare throughout its vast range and is extinct in many regions. Other species have fared even worse: the bush dog and the maned wolf, for example, are on the brink of extinction; the red wolf only survived in zoos but has been reintroduced in the wild. On the other hand, the coyote and the red fox—both opportunists—have benefited from the spread of urban environments and are more abundant than ever before.



STRENGTHENING BONDS

Establishing and maintaining bonds between pack members is essential for the survival of dogs that live in social groups. In African wild dogs, bonding behavior, such as licking and whining, frequently occurs before a hunt. It is only by cooperating and hunting as a team that the dogs are able to bring down and kill prey larger than themselves.